

Clinical Risk Assessment Report: Patient 21

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.41x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.1%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.65 [Bottom 24%] (-2.1%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 81.0 [Top 1%] (-1.4%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.1 [Top 19%] (-1.0%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.07 [Avg (47th %ile)] (-0.9%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.55 [Avg (29th %ile)] (-0.8%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.59 [Avg (30th %ile)] (-0.7%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -1.26 [Bottom 13%] (-0.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.03 [Top 17%] (-0.6%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Machine Learning Feature Attribution (SHAP Decision Path)

